■ Developer Mock Test Results

Student: **Shiva kasula**

Name:	Shiva kasula	**Student ID:**	112
Date:	2026-03-05 12:57:21	**Performance:**	■ Average
Overall Score:	12.0/25 (48.0%)	**Test ID:**	98a50f07-473a-4704...
Warnings:	0/3	**Status:**	Completed ■

■ Section-wise Performance

Section	Score	Percentage	Status
■ Aptitude	4/10	40.0%	■■ Needs Work
■ MCQ/Theory	8/10	80.0%	■ Pass
■ Coding	0.0/5	0.0%	■■ Needs Work

■ Detailed Question Review

■ APTITUDE SECTION (4/10 - 40.0%)

■ Q1. A mixture of two types of tea is to be made, one type costs $5 per kg and the other type costs $10 per kg, if the average cost of the mixture is $7 pe...

Your Answer: 1:2
Correct Answer: 2:1
■ *To find the ratio of the two types of tea, let's denote the amount of the $5 tea as x and the amount of the $10 tea as y. The average cost of the mixture is $7, so the key calculation step is: (5x + 10y) / (x + y) = 7. Solving for x/y, we get 5x + 10y = 7x + 7y, which simplifies to 3y = 2x, or x/y = 3/2. However, the question asks for the ratio of the two types of tea, and since x represents the $5 tea and y represents the $10 tea, the correct ratio is actually 2:1 (since 3/2 can also be represented as 2:1 when considering the parts of the $5 tea to the $10 tea in simplest form) when considering the parts of each type of tea in the simplest*

■ Q2. If a clock shows 3:15, what is the angle between the hour and minute hands? Select the correct answer from the options below.

Your Answer: 30 degrees
Correct Answer: 75 degrees
■ *To find the angle between the hour and minute hands, we need to calculate the position of each hand. The hour hand moves 30 degrees per hour, so at 3:15, it has moved 3 * 30 + 15 * (30/60) = 90 + 7.5 = 97.5 degrees from the 12 o'clock position. The minute hand moves 360 degrees per hour, so at 3:15, it has moved 15 * (360/60) = 90 degrees from the 12 o'clock position. The user's answer of 30 degrees is likely incorrect because they may have only considered the hour hand's position at 3:00 (90 degrees) and not accounted for its additional movement due to the 15 minutes. The correct angle between the hour and minute hands is 97.5 - 90 = 7.5 degrees, but since the minute hand is behind the hour hand, we need*

■ Q3. A clock strikes 5 and it is known that the time from the clock striking 5 to it striking 6 is 65 minutes, what is the error in the clock per day? Sele...

Your Answer: 260 minutes

Correct Answer: 240 minutes

■ *The correct answer is 240 minutes, but let's verify the calculation: if the clock takes 65 minutes to strike from 5 to 6 (instead of the normal 60 minutes), it is slow by 5 minutes per hour. To find the error per day, we multiply this by 24 hours: 5 minutes/hour * 24 hours = 120 minutes/day (fast) if it were ahead, but since it's slow, the clock would lose 120 minutes in a 24-hour period if it were 5 minutes slow per hour, but since the question asks for error and the clock is slow, we should consider the clock is actually gaining 5 minutes every 65 minutes, so it's 60 minutes (normal hour) / 65 minutes (clock hour) = 0.9231, or the clock hour is 1.0769 times slower, so the clock is slow by (1 - 0.9231) * 24*

■ Q4. A product is on sale for 35% off its original price of $120. How much will you pay for the product?

Your Answer: $90

Correct Answer: $78

■ *To find the sale price, calculate 35% of the original price ($120) and subtract it from the original price. The key calculation step is: $120 * 0.35 = $42, then $120 - $42 = $78. The user likely made a mistake by not correctly calculating the discount amount, leading them to choose $90 instead of the correct answer, $78.*

■ Q5. A person is facing north and then turns 90 degrees to the right, then turns 45 degrees to the left, what direction is the person facing now? Select th...

Your Answer: Northeast

■ *Correct! Well done.*

■ Q6. If "APPLE" is coded as "URYYB", what is the code for "BANANA"? Select the correct answer from the options below.

Your Answer: EOBOEO

■ *Excellent! Right answer.*

■ Q7. If 5 workers can build a house in 12 days, how many workers would be required to build the same house in 5 days? Select the correct answer from the op...

Your Answer: 14

Correct Answer: 12

■ *To find the number of workers required to build the house in 5 days, we can use the formula: (number of workers) * (number of days) = constant. Given that 5 workers can build the house in 12 days, we have 5 * 12 = 60 worker-days. To build the same house in 5 days, we need 60 worker-days / 5 days = 12 workers. The user's answer of 14 is likely incorrect due to a miscalculation of the worker-days required. The correct calculation shows that 12 workers are needed to build the house in 5 days.*

■ Q8. If 3 workers can build a wall in 10 days, how many workers would be required to build the same wall in 5 days? Select the correct answer from the opti...

Your Answer: 4

Correct Answer: 6

■ *To determine the number of workers needed to build the wall in 5 days, we can use the formula: (number of workers) * (number of days) = constant. Given that 3 workers can build the wall in 10 days, we have 3 * 10 = 30 worker-days. To build the same wall in 5 days, we need 30 worker-days / 5 days = 6 workers. The user's answer of 4 is likely incorrect because it doesn't account for the full reduction in time, which requires doubling the workforce from 3 to 6 to halve the construction time from 10 to 5 days.*

■ Q9. A mixture of paint and water is in the ratio 3:4. If 3 cups of paint are added to the mixture, what is the new ratio of paint to water?

Your Answer: 6:4

■ *Correct! Good understanding.*

■ Q10. If $1000 is invested at a compound interest rate of 10% per annum, what will be the amount after 2 years? Select the correct answer from the options b...

Your Answer: $1210

■ *Excellent! Right answer.*

■ MCQ SECTION (8/10 - 80.0%)

■ Q1. How can you compile and run a Java program?

Your Answer: Using a code editor or IDE

■ *Correct! Good understanding.*

■ Q2. What is an example of a simple Java program?

Your Answer: All of the above

■ *Correct! Well done.*

■ Q3. What should variable and method names be in Java?

Your Answer: Meaningful

■ *Correct! Well done.*

■ Q4. What is one of the core concepts in Java used to store data values?

Your Answer: Variables

■ *Correct! Good understanding.*

■ Q5. How are variables declared in Java?

Your Answer: Using the data type

■ *Excellent! Right answer.*

■ Q6. What type of block is used to handle exceptions in Java?

Your Answer: All of the above

■ *Excellent! Right answer.*

■ Q7. What is the process of converting one data type to another in Java?

Your Answer: Casting

■ *Excellent! Right answer.*

■ Q8. What are the main parts of a Java program?

Your Answer: Variables, data types, control structures, methods, classes, objects

■ *Correct! Good understanding.*

■ Q9. What is an abstract class in Java?

Your Answer:

```
A class that can be instantiated but cannot be inherited
```

Correct Answer:

```
A class that cannot be instantiated but can be inherited
```

■ *The correct answer is B) A class that cannot be instantiated but can be inherited, because an abstract class in Java is defined as a class that cannot be instantiated on its own and is intended to be inherited by other classes. The user's answer, C) A class that can be instantiated but cannot be inherited, is incorrect because it actually describes a final class in Java, not an abstract class. The user's likely mistake is confusing the properties of abstract and final classes, which have opposite restrictions on instantiation and inheritance.*

■ Q10. What type of error occurs when a program has a syntax mistake?

Your Answer: Compilation error
Correct Answer: Syntax error
■ *The correct answer is "Syntax error" because it specifically refers to mistakes in the source code's syntax, such as missing or mismatched brackets, that prevent the code from being parsed correctly. The user's answer, "Compilation error", is related but broader, as compilation errors can include syntax errors as well as other issues. The user likely mistook the two terms, as compilation errors are often the result of syntax errors, but not all compilation errors are syntax errors.*

■ CODING SECTION (0.0/5 - 0.0%)

■ Q1. Write a Java program that reads an array of integers from stdin and performs certain operations on it using lambda expressions. **Input Format:** - L...

Your Answer:

```
No answer (Skipped)
```

Correct Answer:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        int n = scanner.nextInt();
        int[] array = new int[n];

        for (int i = 0; i < n; i++) {
            array[i] = scanner.nextInt();
        }

        int sum = 0;
        for (int num : array) {
            sum += num;
        }

        double average = (double) sum / n;

        System.out.println("The sum is: " + sum);
        System.out.println("The average is: " + average);

        scanner.close();
    }
}
```

■ *There is no test case failure to analyze, as no test case results or student code were provided. No feedback can be given without this information. Please provide the test case results and student code to receive accurate feedback.*

■ Q2. Write a Java program that defines a class with inheritance. **Input Format:** - Line 1: The input to process **Output Format:** - The processed outp...

Your Answer:

```
No answer (Skipped)
```

Correct Answer:

```
import java.util.Scanner;

class Animal {
    String name;

    Animal(String name) {
        this.name = name;
    }
```

```
    void sound() {
        System.out.println("The animal makes a sound.");
    }
}

class Dog extends Animal {
    Dog(String name) {
        super(name);
    }

    @Override
    void sound() {
        System.out.println("The dog barks.");
    }
}

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        String input = scanner.nextLine();
        Dog dog = new Dog(input);
        dog.sound();
        System.out.println(dog.name);
        scanner.close();
    }
}
```

■ *There is no code submitted to analyze, so it's impossible to determine why the test cases failed. Since no code was provided, it's unclear what the student's implementation was or how it differed from the expected output. Without the actual code, the reason for the test case failure cannot be determined.*

■ Q3. Write a Java program that defines a StudentGrades class with methods to add and calculate grades. **Input Format: - Line 1: The number of grades - ...**

Your Answer:

```
No answer (Skipped)
```

Correct Answer:

```
import java.util.Scanner;

class StudentGrades {
    private double sum;
    private int count;

    public void addGrade(double grade) {
        sum += grade;
        count++;
    }

    public double calculateAverage() {
        return sum / count;
    }
}

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        int numGrades = scanner.nextInt();
        scanner.nextLine(); // Consume newline left-over

        StudentGrades studentGrades = new StudentGrades();
```

```
        String[] grades = scanner.nextLine().split(" ");
        for (String grade : grades) {
            studentGrades.addGrade(Double.parseDouble(grade));
        }

        double average = studentGrades.calculateAverage();
        System.out.println(average);

        scanner.close();
    }
}
```

■ *There is no code to review, so it's impossible to determine why the test cases failed. Since no code was submitted, it's likely that the student didn't provide a solution to calculate the average grade. Without the actual code, it's not possible to provide specific feedback on what went wrong.*

■ Q4. Write a Java program that reads an integer from stdin and prints the Fibonacci series up to that number. **Input Format:** - Line 1: An integer **Ou...

Your Answer:

```
No answer (Skipped)
```

Correct Answer:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        int n = scanner.nextInt();
        int a = 0;
        int b = 1;
        System.out.print(a + " " + b);
        int next = a + b;
        while (next <= n) {
            System.out.print(" " + next);
            a = b;
            b = next;
            next = a + b;
        }
        scanner.close();
    }
}
```

■ *Since no code was submitted, it's impossible to determine the cause of the failure. However, based on the provided example, the expected output is a Fibonacci series up to the input number, which is 10 in this case. The student's code should be reviewed to ensure it correctly generates the Fibonacci series and stops at the correct number.*

■ Q5. Write a Java program that reads a string from stdin and compresses it using run-length encoding. **Input Format:** - Line 1: The string to compress ...

Your Answer:

```
No answer (Skipped)
```

Correct Answer:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        String input = scanner.nextLine();
        StringBuilder compressed = new StringBuilder();
```

```
        int count = 1;
        for (int i = 1; i <= input.length(); i++) {
            if (i < input.length() && input.charAt(i) == input.charAt(i - 1)) {
                count++;
            } else {
                compressed.append(input.charAt(i - 1));
                compressed.append(count);
                count = 1;
            }
        }

        System.out.println(compressed.toString());
        scanner.close();
    }
}
```

■ *Since no student code was submitted, there's no output to compare with the expected result. The test case failure is due to the absence of any code to process the input string. No output was generated to compress the input string "aaaabbbccc" into the expected "a4b3c3" format.*

■ Recommendations

Areas that need improvement:

- **Aptitude:** Practice more logical reasoning, quantitative aptitude, and problem-solving questions.
- **Coding:** Practice more coding problems on platforms like LeetCode or HackerRank.
